Aniruddha Pandey

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This resume is a language model! (Well, a tiny one, ∼83M parameters) Run it on self-aware-resume.vercel.app. I'm in my 4th-semester of master's at Stevens Institute of Technology. I'm fortunate to be advised by Dr. Nikhil Muralidhar. My current research interests are symbolic regression, mechanistic interpretability and agent memory architectures.

Education

Dec 2026 (Expected) — **M.S. in Computer Science** - Stevens Institute of Technology, Hoboken, NJ

Coursework: Statistical Machine Learning, Deep Learning, NLP, Scientific Discovery in ML, Advanced Algorithms; Next-Gen Healthcare Innovators Symposium 2025 — managed panel curation, speaker coordination, event logistics, and student research pitch presentations.

Sep 2021 — **B.Tech in Computer Science and Engineering** - Lovely Professional University, India

Graduate Research

Jan 2025 Present — **Behavioral Fingerprint Detection**: Framework exploiting divergence tokens—sub-semantic artifacts from LLM generation—for model-family attribution across **GPT-4o**, **Claude**, **Gemini**, **Llama**, and **Mistral**; multi-probe pipeline targeting AUROC ≥ 0.90 against humanization attacks. *(In preparation)*

Tri-Mem: Modality-matched memory hierarchies for multi-agent systems; three-layer cognitive architecture mapping semantic memory (sparse-attention KV chunks), episodic memory (visual compression), and declarative memory (vector RAG) with entropy-driven routing. Extends agent memory horizon from ∼30 to 100+ turns at 90% token reduction; evaluated on a custom NovaCorp AuditBench with five agent ablations. *(In preparation)*

NeuralSINDy: Extending **SINDy** with neural function approximators to discover governing equations from noisy, high-dimensional dynamical systems data. *(In preparation)*

Work Experience

Jun 2026 Aug 2026 — **AI Engineer Intern** - Scholera, Hoboken, NJ (Hybrid)

Built multi-stage retrieval with **Pinecone** and **Vertex AI** reranking alongside an automated **RAGAS** eval suite; implemented a computerized adaptive testing (**IRT**) engine with 2PL/3PL Item Response Theory models and **EAP** ability estimation.

Feb 2024 Nov 2024 — **Senior Frontend Engineer** - TaxTaker, Austin, TX (Remote)

Designed an encrypted **localStorage** library for secure client-side data handling.

Oct 2022 Sep 2023 — **Data Visualization Engineer** - Spontivly, St. Petersburg, FL (Remote)

Built interactive dashboards for enterprise clients including **Balenciaga** and the **NFL**.

Sep 2021 Oct 2022 — **Junior Frontend Engineer** - Astro Malaysia, Kuala Lumpur (Remote)

Built an interactive tournament medal table for Commonwealth Games 2022 on stadiumastro.com.

Skills

Math: linear algebra, information theory, bayesian inference, probability theory, convex/stochastic optimization

Modeling: transformer architectures (encoder/decoder, sparse attention, KV caching), tokenizer design (BPE), training pipelines (pretraining, fine-tuning (SFT, GRPO), ablation design), mechanistic interpretability

Systems: Python, PyTorch, vLLM, SGLang, Weights & Biases, LaTeX

Awards

RAISE-26 Finalist | 2022 SIH Judge | 2021 Microsoft Imagine Cup World Finals Runner-Up | 2020 HackMIT Health Tech Winner | 2019 SAP Analytics Cloud Winner